

Fundamentals of Solid State Physics

Optical Properties

Xing Sheng 盛兴

Department of Electronic Engineering
Tsinghua University

xingsheng@tsinghua.edu.cn



Last Class

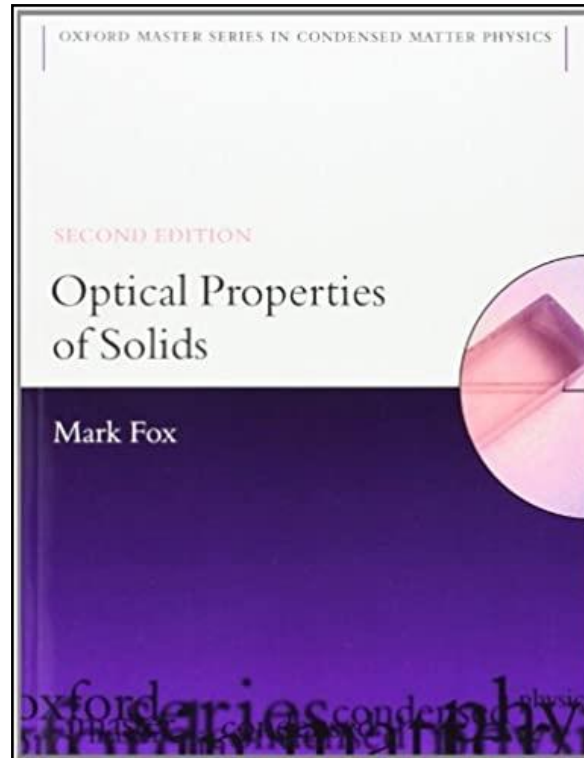
- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- Electronic Properties (Week 4–12)
- **Thermal Properties (Week 13)**
 - ▣ Crystal vibration, phonon band
 - ▣ Thermal conductivity and capacity
- Optical Properties (Week 14)
- Magnetic Properties (Week 15)

This Class

- Introduction (Week 1)
- Materials and Crystal Structures (Week 2–3)
- Electronic Properties (Week 4–12)
- Thermal Properties (Week 13)
- **Optical Properties (Week 14)**
 - ▣ **Origin of Dielectric constant (ϵ) and Refractive index (n)**
 - ▣ **Optical absorption, reflection, refraction, emission**
- Magnetic Properties (Week 15)

Further Reading

- Fox, Chapter 1, 2, 7



Optical Properties of Materials



Metal



SiO₂



Silicon

- **Crystal Structures**
 - polycrystalline, amorphous, single crystalline
- **Electronics**
 - conductor, insulator, semiconductor
- **Optics (in the visible range)**
 - reflective, transparent, absorbing

Fundamentals of Solid State Physics

Optical Processes

Xing Sheng 盛兴

Department of Electronic Engineering
Tsinghua University

xingsheng@tsinghua.edu.cn



Optical Processes

- **Review: Maxwell's Equations**
- **Reflection, Transmission, Absorption, ...**
- **Optical propagation in multi-layers**
 - ▣ **Transfer Matrix Method**

Electrodynamics

- Maxwell's Equations (1865)

$$\nabla \cdot \mathbf{D} = \rho_V$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

$$\oint_s \mathbf{D} \cdot d\mathbf{A} = \int_v \rho \cdot dV$$

$$\oint_s \mathbf{B} \cdot d\mathbf{A} = 0$$

$$\oint_l \mathbf{E} \cdot d\mathbf{l} = -\int_s \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{A}$$

$$\oint_l \mathbf{H} \cdot d\mathbf{l} = \int_s \mathbf{J} \cdot d\mathbf{A} + \int_s \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{A}$$

Electrodynamics

■ Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho_V$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

Constitutive Relations

本构关系

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H}$$

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E}$$

$\varepsilon_0 \varepsilon_r$ - Permittivity (dielectric constant)

$\varepsilon_r = 1$ for vacuum

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

$\mu_0 \mu_r$ - Permeability

$\mu_r = 1$ for vacuum

$\mu_0 = 4\pi \times 10^{-7}$ H/m

Electrodynamics

■ Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho_V$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

Constitutive Relations

本构关系

$$\mathbf{B} = \mu_0 \mu_r \mathbf{H}$$

$$\mathbf{D} = \varepsilon_0 \varepsilon_r \mathbf{E}$$

For most non-magnetic materials (no magnetic field),
 $\mu_r = 1$

Optical properties of materials
is determined by ε_r

Electrodynamics

■ In vacuum

$$\square \rho_V = 0, \mathbf{J} = 0$$

$$\square \mu_r = 1, \epsilon_r = 1$$

$$\nabla \cdot \mathbf{D} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$$

$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$



$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{i(kx - \omega t)}$$

Plane Wave

$$k = \frac{2\pi}{\lambda}$$

$$\omega = \frac{2\pi}{T}$$

wavevector

angular frequency

Electrodynamics

■ In vacuum

$$\square \rho_V = 0, \mathbf{J} = 0$$

$$\square \mu_r = 1, \epsilon_r = 1$$

$$\nabla \cdot \mathbf{D} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$$



$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{i(kx - \omega t)}$$

Plane Wave

$$c = \frac{\omega}{k} = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8 \text{ m/s}$$

light speed in vacuum

Electrodynamics

■ In a dielectric medium

$$\square \rho_V = 0, \mathbf{J} = 0$$

$$\square \mu_r = 1, \epsilon_r \neq 1$$

$$\nabla \cdot \mathbf{D} = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$$



$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \epsilon_r \frac{\partial^2 \mathbf{E}}{\partial t^2}$$

$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{i(kx - \omega t)}$$

Plane Wave

$$v = \frac{\omega}{k} = \frac{c}{\sqrt{\epsilon_r}} = \frac{c}{n}$$

$$\epsilon_r = n^2$$

light speed in a material
 n - refractive index (折射率)

Electrodynamics

- In a dielectric medium

- $\rho_V = 0, \mathbf{J} = 0$

- $\mu_r = 1, \epsilon_r \neq 1$

$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{i(kx - \omega t)}$$

Plane Wave

$$k = \frac{2\pi}{\lambda^*} = \frac{2\pi}{\lambda_0} n$$

λ^* - wavelength in the medium

λ_0 - wavelength in vacuum

Frequency ω does not change

Complex Form of ε_r and n

$$\tilde{\varepsilon}_r = \tilde{n}^2$$

$$\tilde{\varepsilon}_r = \varepsilon_1 + i\varepsilon_2$$

$$\tilde{n} = n + i\kappa$$

$$\rightarrow \begin{cases} \varepsilon_1 = n^2 - \kappa^2 \\ \varepsilon_2 = 2n\kappa \end{cases}$$

ε_r and n depend on optical frequency / wavelength

Complex Form of ε_r and n

$$\begin{cases} n = \frac{1}{\sqrt{2}} \left(\varepsilon_1 + \sqrt{\varepsilon_1^2 + \varepsilon_2^2} \right)^{1/2} \\ \kappa = \frac{1}{\sqrt{2}} \left(-\varepsilon_1 + \sqrt{\varepsilon_1^2 + \varepsilon_2^2} \right)^{1/2} \end{cases}$$

when $\varepsilon_1 \gg \varepsilon_2$ (or $n \gg \kappa$), weakly absorbing

$$\rightarrow \begin{cases} n \approx \sqrt{\varepsilon_1} \\ \kappa \approx \frac{\varepsilon_2}{2n} \end{cases}$$

Reflection 反射

Incident wave

$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{i(kx - \omega t)}$$

Reflective wave

$$\mathbf{E}_R(x, t) = \mathbf{E}_R e^{i(-kx - \omega t)}$$

Reflectivity 反射率

based on boundary conditions
of Maxwell's Equations

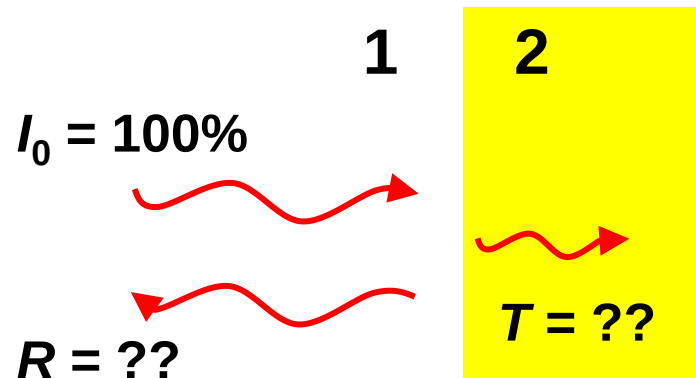
$$R = \left| \frac{\mathbf{E}_R}{\mathbf{E}_0} \right|^2 = \left| \frac{\tilde{n}_2 - \tilde{n}_1}{\tilde{n}_2 + \tilde{n}_1} \right|^2$$

Transmission 透射率

$$T = 1 - R$$

Intensity

$$I \propto |\mathbf{E}|^2$$



If medium 1 is air ($\tilde{n}_1 = 1$)

$$R = \left| \frac{\tilde{n}_2 - 1}{\tilde{n}_2 + 1} \right|^2 = \frac{(n - 1)^2 + \kappa^2}{(n + 1)^2 + \kappa^2}$$

for normal incidence ($\theta = 0$)

Absorption 吸收

Incident wave

$$\mathbf{E}(x, t) = \mathbf{E}_0 e^{i(kx - \omega t)}$$

After traveling a distance L

$$\begin{aligned} \mathbf{E}_T(x, t) &= \mathbf{E}_0 e^{i(kx - \omega t)} e^{ikL} \\ &= \mathbf{E}_0 e^{i(kx - \omega t)} e^{i2\pi\tilde{n}/\lambda * L} \\ &= \mathbf{E}_0 e^{i(kx - \omega t)} e^{i2\pi n/\lambda * L} e^{-2\pi\kappa/\lambda * L} \end{aligned}$$

Lambert Beer's Law

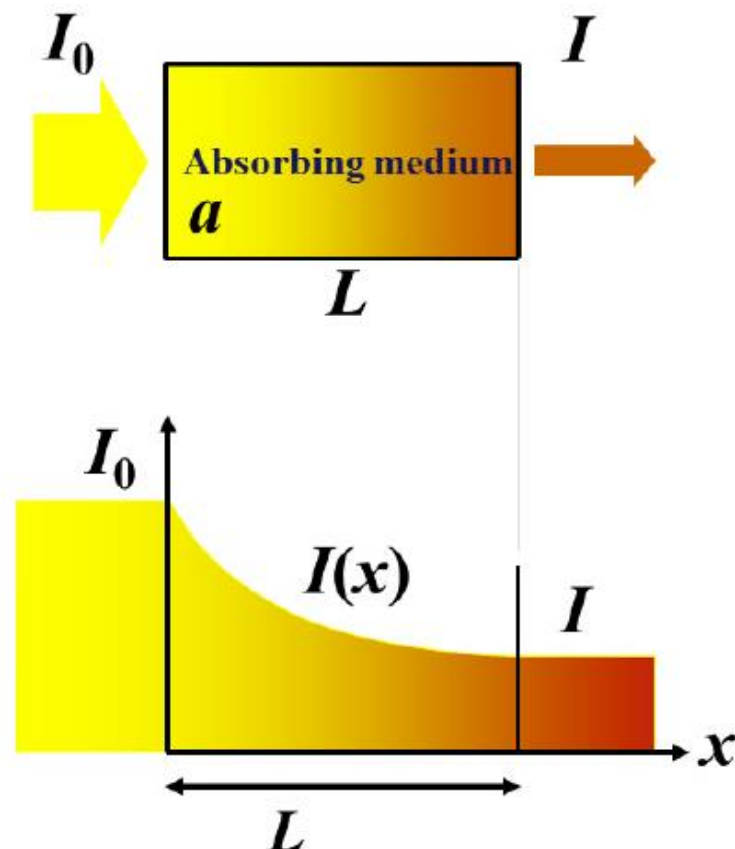
$$I = I_0 e^{-\alpha L}$$

$$\alpha = \frac{4\pi\kappa}{\lambda}$$

absorption coefficient (unit: /m)

Intensity

$$I \propto |\mathbf{E}|^2$$

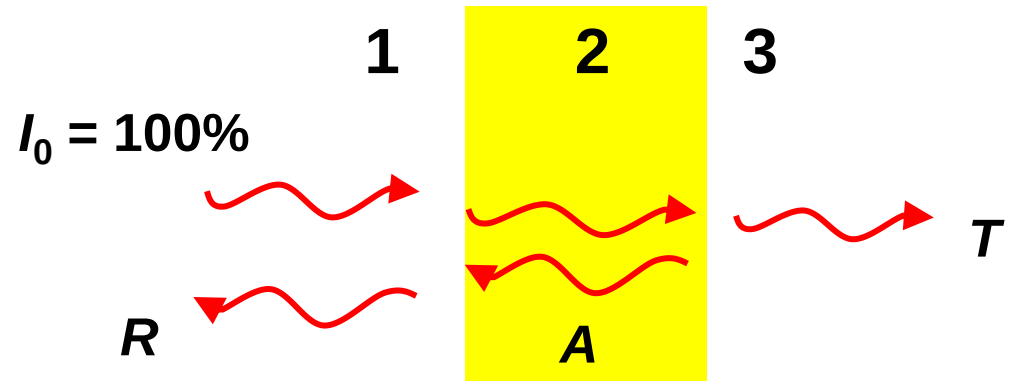


Transmission 透射

Reflection R 反射

Absorption A 吸收

Transmission T 透射



$$R + A + T = 1$$

Example: Silicon

- At $\lambda = 600$ nm, for Si, $\tilde{n} = 3.94 + i*0.025$, calculate
 - ▣ Reflection R at the air/Si interface
 - ▣ Absorption coefficient α at 600 nm
 - ▣ Absorption by a Si film with thickness $L = 0.01$ mm

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} = 35.4\%$$

$$\alpha = \frac{4\pi\kappa}{\lambda} = 5.24 * 10^5 / \text{m}$$

$$A = 1 - e^{-\alpha L} = 99.5\%$$

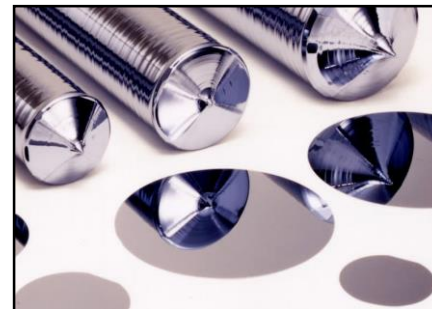
Example: Silicon

- Silicon is a very good absorber at $\lambda = 600 \text{ nm}$
- It can be used to make solar cells and cameras
- Surface reflection is very strong
- It needs an anti-reflective coating ARC (减反膜)

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} = 35.4\%$$

$$\alpha = \frac{4\pi\kappa}{\lambda} = 5.24 * 10^5 / \text{m}$$

$$A = 1 - e^{-\alpha L} = 99.5\%$$



bare
Si
wafer



Si
solar cell
with ARC

Example: Silver

- At $\lambda = 600$ nm, for Ag, $\tilde{n} = 0.12 + i*3.66$, calculate
 - ▣ Reflection R at the air/Ag interface
 - ▣ Absorption coefficient α at 600 nm
 - ▣ Absorption by a Ag film with thickness $L = 100$ nm

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} = 96.7\%$$

$$\alpha = \frac{4\pi\kappa}{\lambda} = 7.67 * 10^7 / \text{m}$$

$$A = 1 - e^{-\alpha L} = 99.95\%$$

Example: Silver

- Ag is a very good mirror at visible wavelengths
- Light can only propagate in Ag at a very small depth

$$R = \frac{(n-1)^2 + \kappa^2}{(n+1)^2 + \kappa^2} = 96.7\%$$

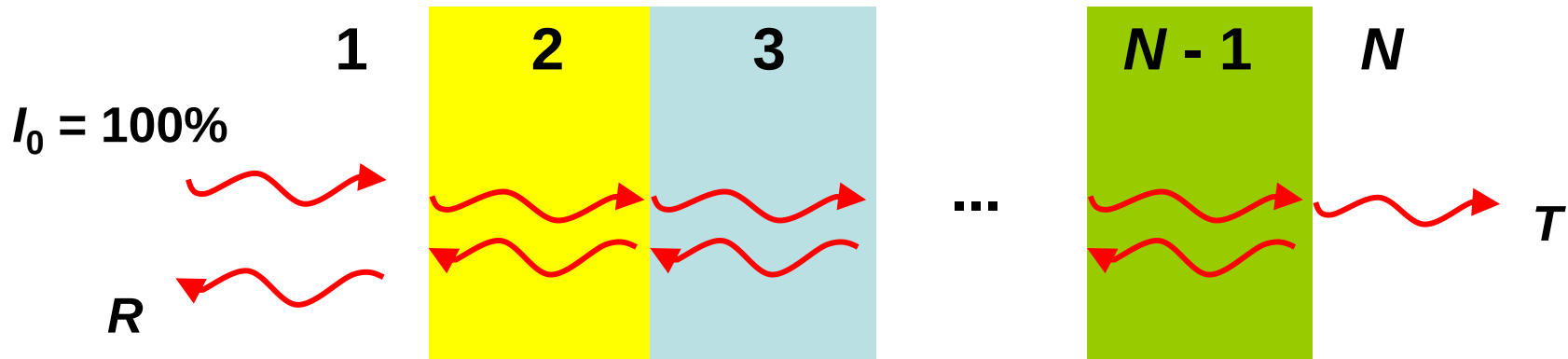
$$\alpha = \frac{4\pi\kappa}{\lambda} = 7.67 * 10^7 / \text{m}$$

$$A = 1 - e^{-\alpha L} = 99.95\%$$



mirror reflection

Multilayer Optical Structures



Solution based on the boundary conditions of Maxwell's Equations

calculated by *Transfer Matrix Method*
Lecture Note 5.2

Example: Anti-Reflective Coating (ARC)

At $\lambda = 600$ nm, no ARC

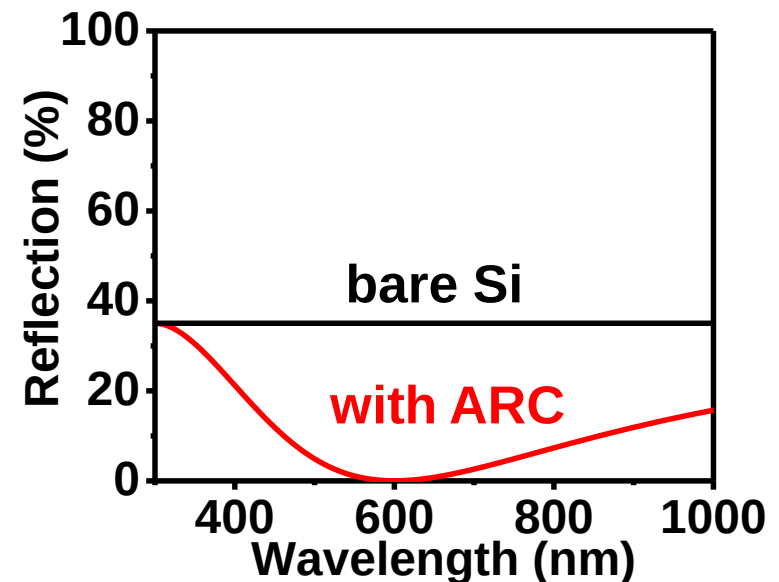
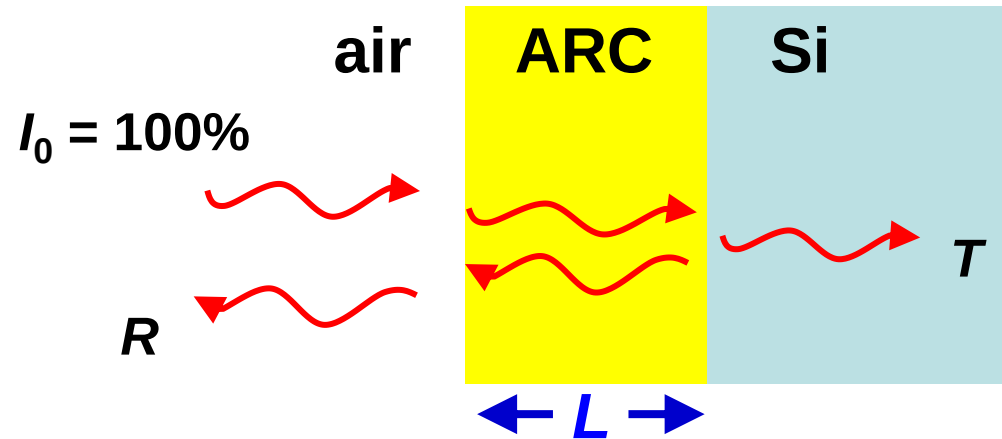
$$R(\text{air/Si}) = 35.4\%$$

Design an ARC

$$n = \sqrt{n(\text{air}) * n(\text{Si})} = 1.98$$

$$L = \frac{\lambda}{4n} = 75 \text{ nm}$$

$$R(\lambda = 600 \text{ nm}) = 0$$



Homework 9.2

Example: ARC for Si

At $\lambda = 600$ nm, no ARC

$$R(\text{air/Si}) = 35.4\%$$

Design an ARC

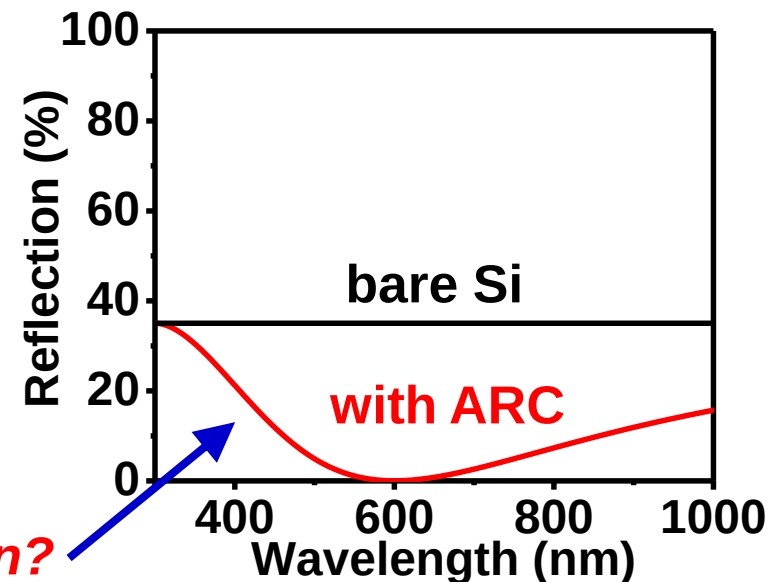
$$n = \sqrt{n(\text{air}) * n(\text{Si})} = 1.98$$

$$L = \frac{\lambda}{4n} = 75 \text{ nm}$$

$$R(\lambda = 600 \text{ nm}) = 0$$



A Si solar cell
with ARC
looks blue



Q: How to further reduce the reflection?

Example: ARC for Glass

For glass

$$n = 1.45$$

At $\lambda = 600$ nm, no ARC

$$R(\text{air/glass}) = 3.4\%$$

Design an ARC

$$n = \sqrt{n(\text{air}) * n(\text{glass})} = 1.2$$

$$\text{thickness} = \frac{\lambda}{4n} = 125 \text{ nm}$$

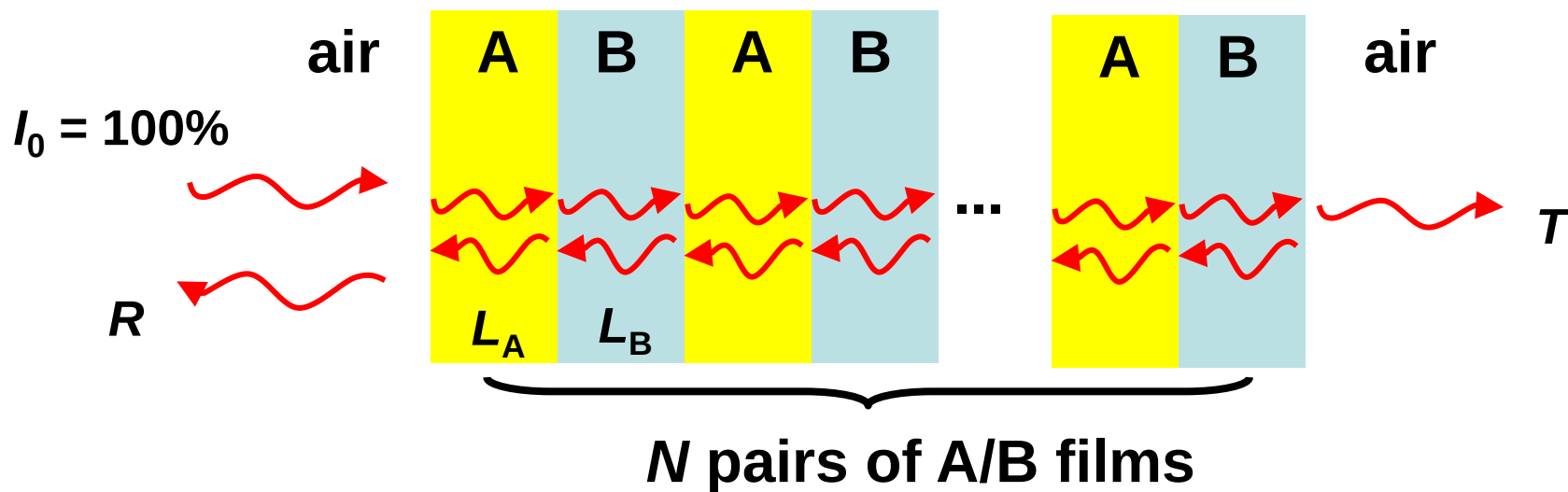


without ARC



with ARC

Example: Bragg Reflector



If we choose

$$L_A = \frac{\lambda}{4n_A}$$

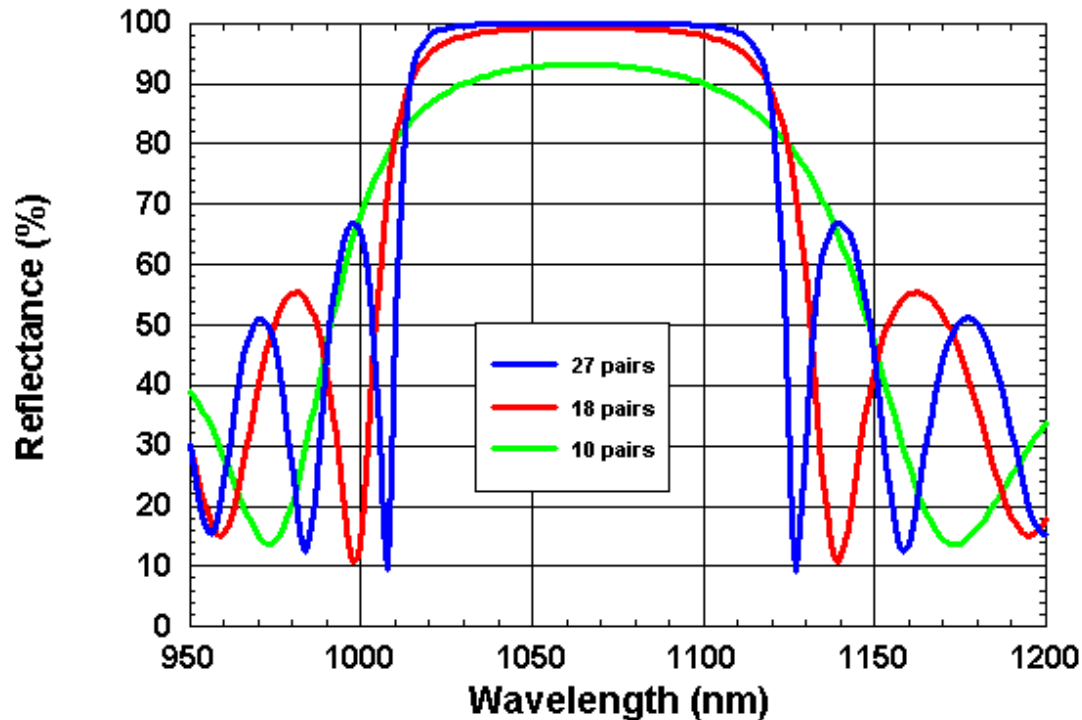
$$L_B = \frac{\lambda}{4n_B}$$

Project 3

$$R = \left(\frac{n_A^{2N} - n_B^{2N}}{n_A^{2N} + n_B^{2N}} \right)^2$$

If $n_A \neq n_B$, when $N \rightarrow +\infty$
 $R \rightarrow 100\%$

Example: Bragg Reflector



**A perfect mirror
(Bragg mirror)
(better than silver)**

**1D photonic crystal
(光子晶体)**

Project 3

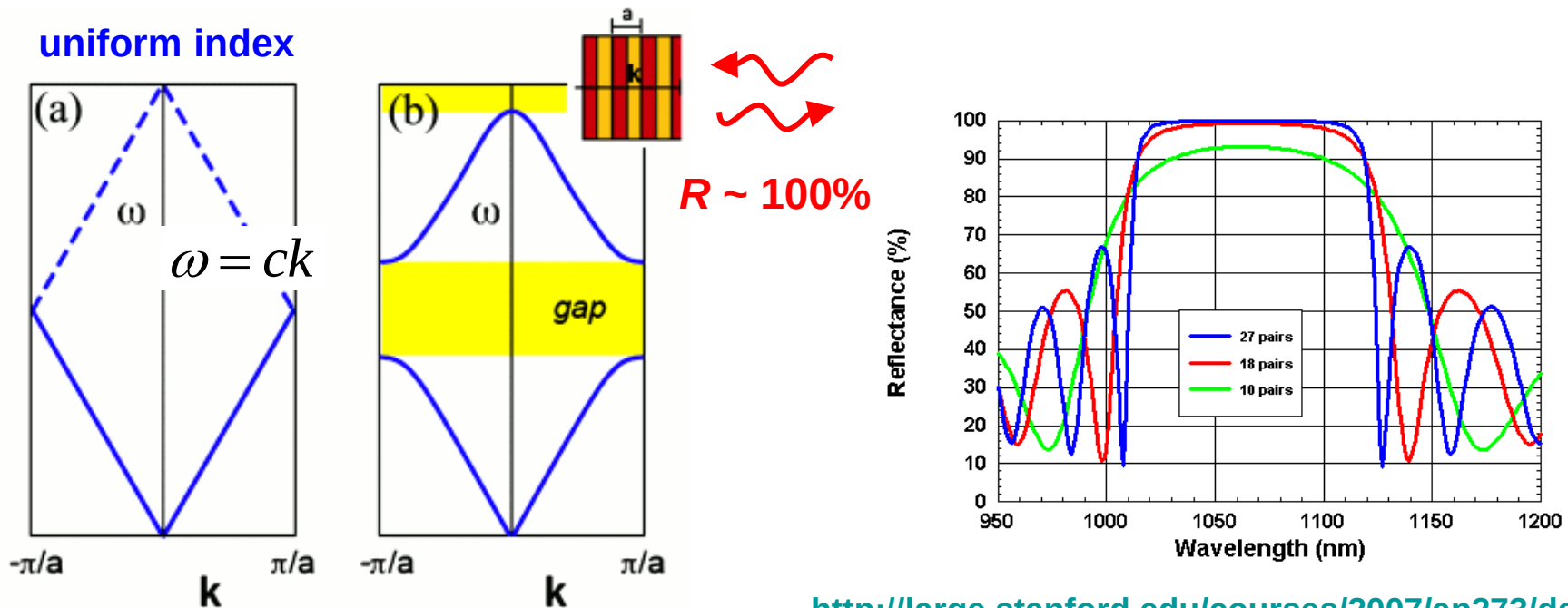
$$R = \left(\frac{n_A^{2N} - n_B^{2N}}{n_A^{2N} + n_B^{2N}} \right)^2$$

If $n_A \neq n_B$, when $N \rightarrow +\infty$

$R \rightarrow 100\%$

Photonic Crystals (光子晶体)

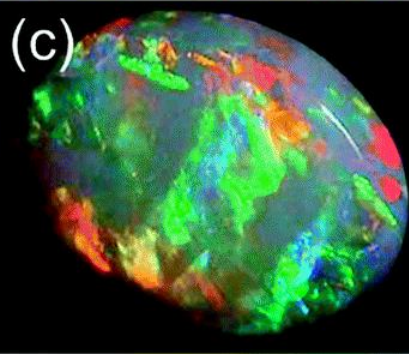
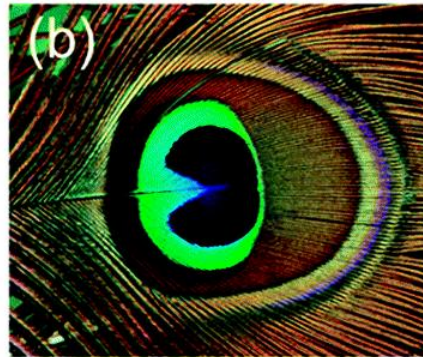
- Periodically structured optical media
 - Forming photonic band gaps
 - no light can pass through (~100% reflection)
 - color created by structure, not material absorption



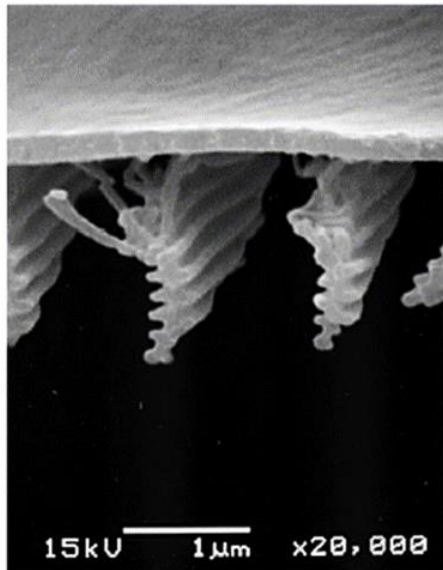
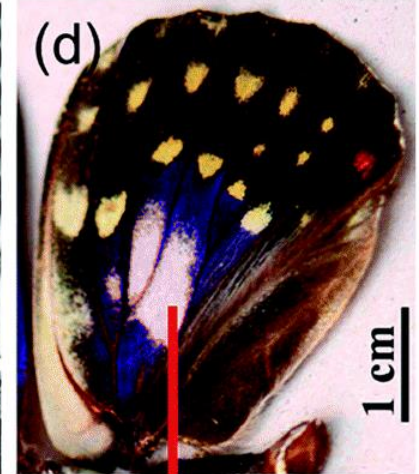
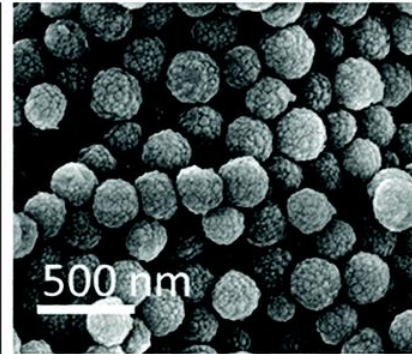
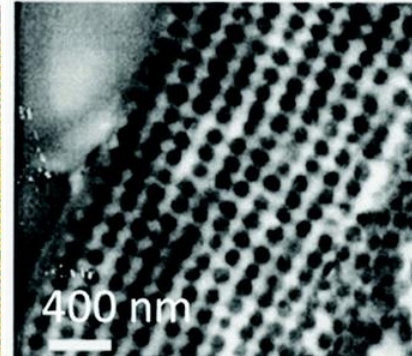
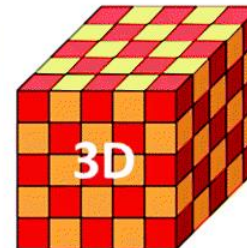
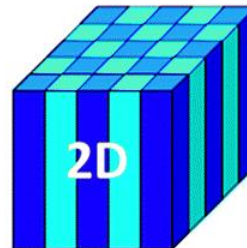
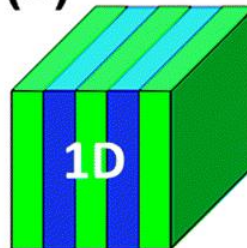
Photonic Crystals in Nature



(a)



(e)



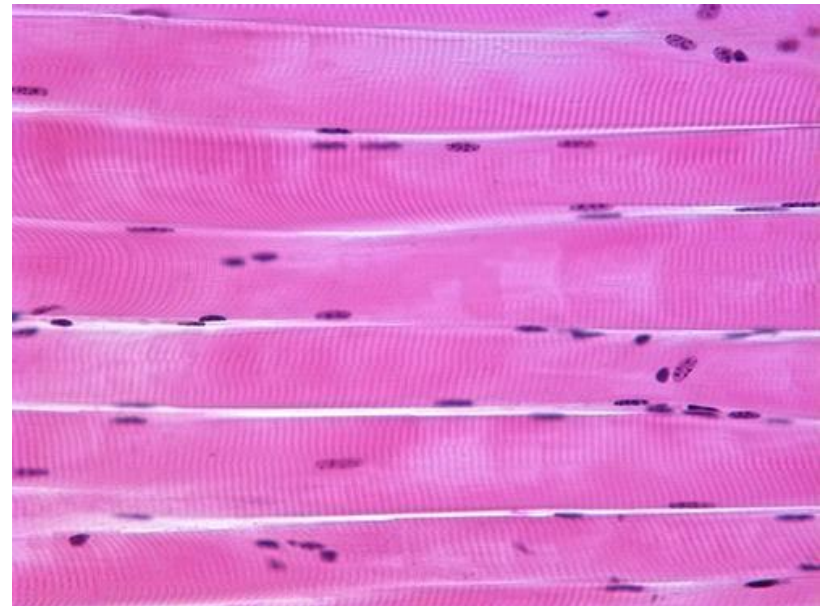
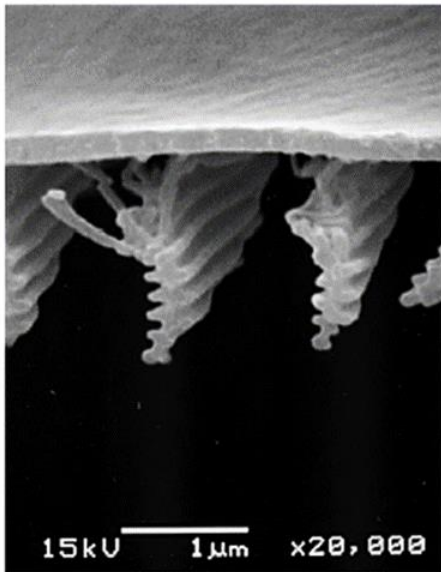
Photonic Crystals in Nature



(a)



**structure color
of beef**



Thank you for your attention